

Aspartate Aminotransferase (AST/SGOT)

Test Details

METHODOLOGY

Kinetic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

A wide range of disease entities alters AST (SGOT), with origin from many organs. When an increased AST is from the liver, it is more likely to relate to disease of the hepatocyte. Other enzymes, including alkaline phosphatase and GT, are more sensitive indicators of biliary obstruction.

Causes of low AST: uremia, vitamin B₆ deficiency (this can be corrected), metronidazole, trifluoperazine.

Causes of high AST: chronic alcohol ingestion, not limited to overt chronic alcoholism; cirrhosis. In alcoholic hepatitis, AST values usually are <300 units/L. In hepatitis, look for a high AST:LD (LDH) ratio, >3, and very high AST peaking at 500–3000 units/L in acute viral hepatitis (ie, in clinical acute viral hepatitis the transaminases may be increased 10 times or more above their upper limits of normal). AST increases are found in other types of liver disease, including earlier stages of hemochromatosis; chemical injury (eg, necrosis related to toxins such as carbon tetrachloride). Some instances of cholecystitis cause increased AST.

AST and ALT (SGPT) are increased in Reye syndrome.^{1,2} In infectious mononucleosis, LD (LDH) is commonly considerably higher than AST. Trauma (including head trauma and including surgery) and other striated muscle diseases, including dystrophy, dermatomyositis, trichinosis, polymyositis, and gangrene cause AST increases. Both AST and ALT elevations are found with Duchenne muscular dystrophy. Look for high CK in myositis, high LD₅ (or isomorphic pattern in some instances of polymyositis) on LD isoenzymes.

In myocardial infarction AST peaks about 24 hours after infarct and returns to normal three to seven days later. In acute MI without shock or heart failure, ALT is not apt to increase significantly. AST increases in congestive failure with centrilobular liver congestion, in which high LD₅ on LD isoenzymes is found, and in pericarditis, myocarditis, pancreatitis, and other inflammatory states including Legionnaires' disease. In renal infarction LD is usually high, out of proportion to AST.³ Lung infarction and other disease entities leading to necrosis including large, necrotic tumors cause increased AST; LD is commonly also increased in such instances. Shock (LD also usually increased); hypothyroidism (LD and/or CK not infrequently increased in myxedema); hemolytic anemias (LD high with increased LD₁) and certain CNS diseases may increase AST.

Very high AST levels usually are caused by liver disease and/or by shock.

Drugs: A large number of commonly used drugs have been reported to elevate AST: isoniazid, phenothiazines, erythromycin, progesterone, anabolic-androgenic steroids, halothane, methyldopa, opiates, indomethacin, salicylates in children, and other drugs. Hepatotoxicity from drugs may cause high aminotransferase activity with elevation of AST:ALT ratio.⁴

Acetaminophen hepatotoxicity deserves special mention. In alcoholics, apparently moderate doses of the analgesic have caused severe hepatotoxicity. Doses of 2.6–16.5 g/24 hours are reported with total bilirubin 1.3–23.9 mg/dL, AST 1960–29,700 units/L, and ALT 12,000–12,550 units/L. The characteristic pattern included mild to severe coagulopathy and AST greater than ALT by a considerable margin.⁵

Macroenzyme causing unexplained increase of AST is described with normal levels of CK and ALT.⁶

Custom Additional Information

AST has origin from heart, liver, skeletal muscle, kidney, pancreas, spleen, and lung. Very high values, >500 units/L, usually suggest hepatitis or other kinds of hepatocellular necrosis but can also be found with large necrotic tumors, other types of necrosis or extensive hypoxia, congestive failure, and shock. Unexplained AST elevations should first be investigated with ALT and GT. Mitochondrial AST (m-AST) may be useful in the diagnosis of alcoholic liver disease; it is reviewed by Rej.⁴

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Gross hemolysis; improper labeling

References

Tonks DB. A study of the accuracy and precision of clinical chemistry determinations in 170 Canadian laboratories. *Clin Chem*. 1963 Apr; 9:217-233. PubMed 13985504 (<http://www.ncbi.nlm.nih.gov/pubmed/13985504>)

Footnotes

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2. DeVivo DC. How common is Reye's syndrome? *N Engl J Med*. 1983 Jul 21; 309(3):179-181. PubMed 6866017 (<http://www.ncbi.nlm.nih.gov/pubmed/6866017>)
3. Winzelberg GG, Hull JD, Agar JW, Rose BD, Pletka PG. Elevation of serum lactate dehydrogenase levels in renal infarction. *JAMA*. 1979 Jul 20; 242(3):268-269. PubMed 448917 (<http://www.ncbi.nlm.nih.gov/pubmed/448917>)
4. Rej R. Aminotransferase in disease. *Clin Lab Med*. 1989 Dec; 9(4):667-687. PubMed 2686908 (<http://www.ncbi.nlm.nih.gov/pubmed/2686908>)
5. Seeff LB, Cuccherini BA, Zimmerman HJ, Adler E, Benjamin SB. Acetaminophen hepatotoxicity in alcoholics. A therapeutic misadventure. *Ann Intern Med*. 1986 Mar; 104(3):399-404 (review). PubMed 3511825 (<http://www.ncbi.nlm.nih.gov/pubmed/3511825>)
6. Litin SC, O'Brien JF, Pruett S, et al. Macroenzyme as a cause of unexplained elevation of aspartate aminotransferase. *Mayo Clin Proc*. 1987 Aug; 62(8):681-687. PubMed 3600038 (<http://www.ncbi.nlm.nih.gov/pubmed/3600038>)